

Filtered Approximate Nearest Neighbor Search on Vectors with Diverse Labels

1 PSEUDO CODE OF ApproximateELS

The approximated ELS retrieval method (ApproximateELS, Algorithm 1) initially checks whether $L_q \in F$ (Lines 2-6). If $L_q \in F$, it returns $\hat{F}_q = \{L_q\}$ (Line 8). Otherwise, the algorithm enumerates the nodes in $\mathbb{I}(l_{\max})$, where $l_{\max}$ is the queried label with the largest ID (Lines 9-10). Each such node is treated as a possible anchor on a trie path that may contain all labels in L_q . The function BOTTOMUP-TRAVERSAL then walks from the anchor toward the root and matches query labels in reverse ID order (Lines 12-21). During this traversal, ancestor labels with IDs larger than the current target label are skipped, a matched label advances the query pointer, and the traversal is pruned once a smaller label or the root is reached before all query labels are matched (Lines 16-21). If all query labels are matched, the anchor is regarded as a valid starting point for relaxed ELSs (Lines 22-23).

Finally, COLLECTCANDIDATEELSs performs a breadth-first traversal from the validated anchor node and inserts every encountered terminal label set into $\hat{F}_q^{(a)}$ (Lines 24-32). Unlike exact ELS retrieval, ApproximateELS does not further filter these candidates to retain only minimal supersets of L_q . Therefore, the returned set may contain non-minimal supersets, but each collected terminal label set remains compatible with the query-label containment condition.

2 THEOREM AND PROOF

THEOREM 1. *For every label set $L \in \hat{F}_q^{(a)}$ such that $\hat{F}_q \subseteq \hat{F}_q^{(a)} \subseteq F_{\text{pass}}$, by employing at least one entry vector whose associated label set is L , the traversal on the unified graph can reach all qualified vectors without encountering any unqualified ones.*

PROOF. First, for any $S \subseteq F$, we use $\text{Desc}^+(S)$ to represent the group of label sets that can be reached by a traversal of LNG starting from nodes that correspond to label sets inside S . Since $\hat{F}_q$ is the set of minimum supersets of L_q on LNG, we have

$$\text{Desc}^+(\hat{F}_q) = F_{\text{pass}}. \quad (1)$$

Moreover, since $\hat{F}_q \subseteq \hat{F}_q^{(a)} \subseteq F_{\text{pass}}$, we have

$$\text{Desc}^+(\hat{F}_q) \subseteq \text{Desc}^+(\hat{F}_q^{(a)}) \subseteq \text{Desc}^+(F_{\text{pass}}). \quad (2)$$

Furthermore, given that edges on LNG indicate label set containment relations, we have

$$\text{Desc}^+(F_{\text{pass}}) = F_{\text{pass}}. \quad (3)$$

The above equations indicate that

$$\text{Desc}^+(\hat{F}_q^{(a)}) = F_{\text{pass}}. \quad (4)$$

That is to say, a traversal of LNG starting from nodes that correspond to label sets inside $\hat{F}_q^{(a)}$ reaches the exact set of qualified label sets. Further consider a traversal of the unified graph starting from at least one entry vector whose associated label set is L . The combination of cross-group and intra-group proximity edges enables this traversal to exactly reach all vectors whose associated label set is inside $\text{Desc}^+(\{L\})$. As a result, Equation (4) indicates that, the traversal on the unified graph can reach all qualified vectors without encountering any unqualified ones, by employing at least

Algorithm 1 ApproximateELS

Input: trie tree T with root r , inverted list $\mathbb{I}$, and the queried label set L_q

Output: the set $\hat{F}_q^{(a)}$ of approximated ELSs

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1:  $\hat{F}_q^{(a)} \leftarrow \emptyset$ , current node  $c \leftarrow \text{root } r$  of  $T$ 
2: for each queried label  $l \in L_q$  in the increasing order of IDs do
3:   if no child of  $c$  corresponds to  $l$  then
4:     break
5:   else
6:      $c \leftarrow$  child of  $c$  that corresponds to  $l$ 
7: if  $c$  is a terminal node that corresponds to  $L_q$  then
8:   return  $\hat{F}_q^{(a)} = \{L_q\}$ 
9: for each node  $u \in \mathbb{I}(l_{\max})$  do
10:  BOTTOMUP-TRAVERSAL( $L_q, u$ )
11: return  $\hat{F}_q^{(a)}$ 
12: function BOTTOMUP-TRAVERSAL( $L_q, u$ )
13:   $u_{\text{initial}} \leftarrow u, i \leftarrow |L_q| - 1$ 
14:  while  $i \geq 0$  do
15:     $u_p \leftarrow$  parent of  $u$  on the trie tree
16:    if  $u_p$  matches a label with a larger ID than  $L_q[i]$  then
17:       $u \leftarrow u_p$ 
18:    else if  $u_p$  matches  $L_q[i]$  then
19:       $u \leftarrow u_p, i \leftarrow i - 1$ 
20:    else if  $u_p$  matches a label with a smaller ID than  $L_q[i]$  or  $u_p$  is  $r$  then
21:      break
22:    if  $i < 0$  then
23:      COLLECTCANDIDATEELSs( $u_{\text{initial}}$ )
24: function COLLECTCANDIDATEELSs( $u_i$ )
25:   $Q \leftarrow \{u_i\}$ 
26:  while  $Q \neq \emptyset$  do
27:     $v \leftarrow Q.\text{dequeue}()$ 
28:    if  $v$  is a terminal node corresponding to  $L \in F$  then
29:       $\hat{F}_q^{(a)}.\text{insert}(L)$ 
30:    else
31:      for each child  $x$  of  $v$  on  $T$  do
32:         $Q.\text{enqueue}(x)$ 

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one entry vector whose associated label set is L , for every $L \in \hat{F}_q^{(a)}$. Hence, this theorem holds. $\square$